

6.3 Types of Data (Qualitative vs Quantitative)

کیفیتی ڈیٹا (Qualitative Data) الفاظ پر مشتمل ہوتا ہے اور بتاتا ہے کہ لوگ کیا سوچتے، محسوس کرتے یا تجربہ کرتے...

CS XI

Chapter 6: Digital Literacy

Lecture 02 of 14

★ Golden

6.3 Types of Data (Qualitative vs Quantitative) ★ Golden

LEARN IT

Data means facts, numbers, words or observations that we collect. The data gathered through a survey, an investigation or a classroom project can be described in two ways: **qualitatively** or **quantitatively**. Knowing the difference is critical, because each type is analysed in a different way.

6.3.1 Qualitative Data

Qualitative data describes what people **think, feel, notice or experience**. It is **non-numerical** and **descriptive**: it is made of words, not numbers. The word comes from *quality* (what something is like).

Examples: students' opinions about online classes; descriptions of study habits; observation notes ("The students looked bored after 20 minutes").

6.3.2 Quantitative Data

Quantitative data deals with **numbers**. The word comes from *quantity* (how much or how many). It is useful when you need to **count, measure, compare, calculate averages**, or **display results in charts**.

Examples: number of students using the internet daily; marks obtained in a test; time spent online in hours.

6.3.3 Comparing Qualitative and Quantitative Data

Type of data	Nature	Examples	Best used for
Qualitative	Descriptive, non-numerical	Opinions, reasons, behaviour	Understanding why and how something happens
Quantitative	Numerical, countable or measurable	Hours, scores, percentages, totals	Comparing results and making graphs/charts

LEARN IT

Going a little further (background). Quantitative data is sometimes split into two kinds:

- **Discrete** data can only take separate, countable values, e.g. number of students in a class (35, 36 – never 35.5).
- **Continuous** data can take any value in a range and is *measured*, e.g. height (162.4 cm) or time online (2.75 hours).

Also note: a question with fixed categories (e.g. "Which activity do you use your phone for most? Lectures / Notes / Social media / Games") gives **categories** that you can **count**. In this chapter's case study such answers are treated as **quantitative (categories)**, because you count how many students chose each option.

WORKED EXAMPLE

Classify each piece of data.

Data	Type	Reason
"I like online classes because I can replay lectures."	Qualitative	An opinion with a reason, in words.
Ayesha scored 78 out of 100.	Quantitative	A score – a number you can compare.
28 out of 40 students use the internet daily.	Quantitative	A count (and gives 70%).
Observer notes: "Students chat during the last period."	Qualitative	A description of behaviour.
Bilal spends 3.5 hours online per day.	Quantitative	A measurement of time.

Quick test: ask "Can I calculate an average of this?" If yes, it is quantitative. If the answer is words, it is qualitative.

REMEMBER IT

QUALitative = **QUAL**ity = words and feelings. **QUANT**itative = **QUANT**ity = numbers and counts. Qualitative answers **why/how**; quantitative answers **how many/how much**.

COMMON MISTAKE

Mistake: Thinking that any data containing a number is quantitative. A phone number or a roll number is a *label*, not a quantity – averaging roll numbers is meaningless. **Fix:** Data is quantitative only if the number measures or counts something.

کیفیتی ڈیٹا (Qualitative Data) الفاظ پر مشتمل ہوتا ہے اور بتاتا ہے کہ لوگ کیا سوچتے، محسوس کرتے یا تجربہ کرتے ہیں، جیسے آن لائن کلاسز کے بارے میں رائے۔ مقداری ڈیٹا (Quantitative Data) اعداد پر مشتمل ہوتا ہے، جیسے ٹیسٹ کے نمبر یا آن لائن گزارے گئے گھنٹے۔ کیفیتی ڈیٹا "کیوں اور کیسے" سمجھنے کے لیے، جبکہ مقداری ڈیٹا موازنہ کرنے اور چارٹ بنانے کے لیے بہترین ہے۔

EXAM TIP

Common questions: "Differentiate between qualitative and quantitative data" (4–5 marks) or "Classify the following data". Write the answer as a **table** with four rows: **nature**, **examples**, **best used for**, plus a one-line definition of each. Always give **two examples** of each type. The source's table (Nature / Examples / Best used for) is exactly what examiners expect.

CHECK YOURSELF

1. Is "time spent online in hours" qualitative or quantitative?
2. Which type of data is best for understanding *why* something happens?
3. Classify: "The canteen food is too oily."

Answers: 1. Quantitative (measurable number). 2. Qualitative. 3. Qualitative (an opinion in words).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Qualitative data is:

- a. Categories / words (favourite subject)
- b. Heights in cm only
- c. Always continuous numbers
- d. Always bits

MCQ 2. Quantitative data is:

- a. Numeric measurements or counts
- b. Colours of bags only
- c. Always opinions
- d. Always URLs

MCQ 3. Favourite colour is:

- a. Qualitative
- b. Quantitative continuous
- c. A primary key always
- d. A gate

MCQ 4. Height is:

- a. Quantitative continuous
- b. Qualitative
- c. Boolean only
- d. Tertiary

MCQ 5. Number of siblings is:

- a. Quantitative discrete
- b. Qualitative
- c. Analog voltage
- d. A query

MCQ 6. Why the type matters:

- a. It decides the chart and statistic
- b. It decides OSI layer
- c. It decides NAND grouping
- d. It decides Agile sprint length

Extra short questions

1. Define qualitative vs quantitative with examples.
2. Discrete vs continuous quantitative.
3. Which chart for qualitative shares?
4. Which average for qualitative?

Extra long question

1. A survey has: gender, hours of study, favourite app, test marks. Classify each, pick a chart, and say mean/median/mode which is legal.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Qual: labels (blood group). Quant: numbers (marks).

2. Discrete: counts (students). Continuous: measurements (time, temperature).
3. Bar or pie (few categories).
4. Mode — mean of “red/blue” is meaningless.

Long

1. Gender qual → pie/bar, mode. Hours quant cont → hist, mean/median. App qual → bar, mode. Marks quant → hist/box, mean/median. Table.

Next lecture: 6.4 Data-Collection Strategies